

IT 220 – Business Data Communications

Fall 2024 Final Course Project

Network Design Project

Student Name: Christos Paloubis

Course Instructor: [Instructor Name]

Submission Date: December 18, 2024

Organizational Overview

This network design project is for a fictitious organization named Tech Solutions Inc., a medium-sized IT consulting firm with a head office and a branch office. The organization requires a reliable and scalable network to support its day-to-day operations, including file sharing, communication, and secure internet access. Each office includes multiple departments such as IT, HR, and Finance, each requiring isolated but interconnected networks. Additionally, the organization supports remote work and requires wireless access for both employees and guests.

Project Description/Abstract

The purpose of this project is to design a functional network for Tech Solutions Inc. using Cisco Packet Tracer. The network will include three individual LANs, one WAN connection, and 15+ end devices. It will use static routing for internal communication, supplemented with RIP or OSPF protocols to enable dynamic routing. The design will incorporate Class A, B, and C IP networks, along with VLANs for efficient segmentation. Wireless access will be provided for all LANs, and subnetting will be implemented to optimize IP address allocation. The network will ensure scalability, security, and reliability for the organization's operations.